ECEN 4/5632 Theory and Application of Digital Filters
Lecture 2, 8-25-26, 8-27-26

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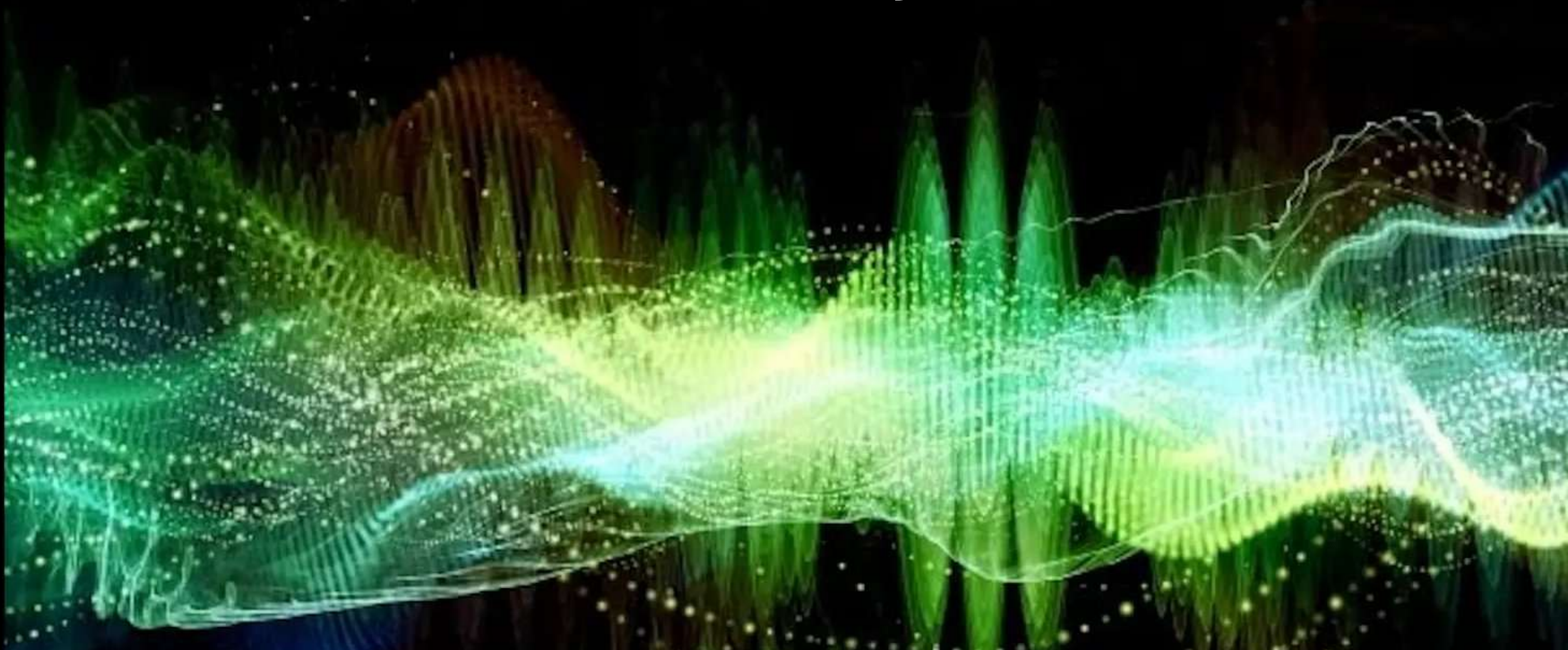


* With generous help from Gemini Pro and ChatGPT

DSP + ML

Eigenfunctions and the DTFT

Week 1 • Lecture 2 • 75 minutes

Why sinusoids survive LTI systems—and why frequency-domain multiplication is so powerful

e^{jωn}
probe

❯

LTI system
h[n]

❯

H(e^{jω})
eigenvalue

❯

DTFT
spectrum

❯

Features
spectrogram
STFT

Today’s learning objectives

00–10 MIN

Move from time-domain mechanics to frequency-domain structure.

- **Prove that complex exponentials are eigenfunctions of LTI systems.**
- Define the DTFT and inverse DTFT and explain 2π periodicity.
- Use time shift, convolution, and Parseval properties.
- Derive frequency response directly from a linear constant-coefficient difference equation.
- Interpret a one-pole filter in time and frequency domains.

Graduate expectation

Be able to state convergence or stability assumptions when a transform/frequency response is used.

00–10 MIN

Recap: an LTI system is convolution by h[n]

Now choose an input that makes convolution simplify dramatically.

$$y[n] = \sum_k h[k]x[n-k]$$

$$x[n] = e^{j\omega n}$$

$$y[n] = \sum_k h[k]e^{j\omega(n-k)}$$

$$= e^{j\omega n}\sum_k h[k]e^{-j\omega k}$$

Notice

The input waveform e^{jωn} factors back out. The system can only multiply it by a complex number.

Complex exponentials are eigenfunctions of every LTI system

00–10 MIN

The eigenvalue depends on frequency.

$$y[n] = H(e^{j\omega})e^{j\omega n}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

What stays the same

- Frequency ω
- Exponential form
- Infinite-duration mode

What changes

- Magnitude: |H(e^{jω})|
- Phase: ∠H(e^{jω})
- Possibly complete cancellation

Frequency response is a gain-and-phase lookup table

10–35 MIN

One complex number for each normalized frequency ω.

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$$

Tone in	Magnitude	Phase	Tone out
$\cos(\omega_0 n)$	× |H(e^{jω$_0$})|	+ ∠H(e^{jω$_0$})	same ω_0

Audio EQ

A graphic equalizer is approximately a deliberately shaped |H(e^{jω})|.

The DTFT analyzes a sequence as a continuous function of frequency

10–35 MIN

Time is discrete; frequency remains continuous.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}\,d\omega$$

Forward DTFT

- Sequence x[n] → function X(e^{jω})
- Weighted sum of rotating phasors
- Requires convergence in an appropriate sense

Inverse DTFT

- Integrates one period of spectrum
- Reconstructs each integer sample
- Any interval of length 2π works

Why is the DTFT 2π-periodic?

10–35 MIN

Because sampled-time exponentials cannot distinguish ω from ω+2πk.

$$e^{-j(\omega+2\pi)n} = e^{-j\omega n} e^{-j2\pi n} = e^{-j\omega n}$$

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

same spectral content repeats every 2π

Normalized frequency ↔ physical frequency

10–35 MIN

Week 2 will make this relationship central.

$$\omega = 2\pi \frac{f}{f_s} \quad \Longleftrightarrow \quad f = \frac{\omega}{2\pi} f_s$$

$$\omega = \pi \quad \Longleftrightarrow \quad f = \frac{f_s}{2}$$

Example

At f_s=48 kHz: $\omega=\pi/2$ ↔ 12 kHz, and $\omega=\pi$ ↔ 24 kHz.

- The interval $-\pi \le \omega < \pi$ represents one unique frequency period.
- Frequencies outside that interval wrap into it modulo 2π.

Two DTFT pairs worth knowing immediately

10–35 MIN

Use them as anchors—not as a substitute for reasoning.

$$\delta[n] \longleftrightarrow 1$$

$$a^n u[n] \longleftrightarrow \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1$$

$$\delta[n - n_0] \longleftrightarrow e^{-j\omega n_0}$$

Why the condition?

For |a|<1, the geometric series is absolutely summable; |a|≥1 does not converge on the unit circle in the ordinary sense.

Time shifting becomes linear phase

35–55 MIN

Delays are easy to recognize in the frequency domain.

$$x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$$

Magnitude

- |e^{-jωn_0}|=1
- A pure delay does not change spectral magnitude

Phase

- Adds $-\omega n_0$ radians
- Linear phase slope encodes delay

Audio connection

An ideal sample delay preserves the spectrum's magnitude exactly; it only rotates phase versus frequency.

Convolution in time ↔ multiplication in frequency

35–55 MIN

This is one of the most important DSP simplifications.

$$y[n] = x[n] * h[n] \qquad\qquad Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

Time domain	DTFT	Frequency domain	IDTFT
slide / multiply / sum	transform	pointwise multiply	reconstruct

Computational echo

FFT-based convolution exploits the same multiplication idea numerically for long signals.

Parseval: energy is preserved across representations

35–55 MIN

The spectrum redistributes energy; it does not create it.

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

Time-domain view

- Energy = sum of squared samples
- Natural for waveforms

Frequency-domain view

- Energy density across ω
- Natural for bands and features

ML bridge

Many audio features—band energies, log spectra, mel features—summarize how signal energy is distributed across frequency.

Difference equations give H(e^{jω}) directly

35–55 MIN

Use the eigenfunction or DTFT property instead of solving every input separately.

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{m=0}^{M} b_m x[n-m] H(e^{j\omega}) = \frac{\sum_{m=0}^{M} b_m e^{-j\omega m}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

- Substitute x[n]=e^{jωn}, y[n]=H(e^{jω})e^{jωn}.
- Factor out e^{jωn}.
- Solve the remaining algebraic equation for H(e^{jω}).

One-pole example: y[n]=0.5y[n−1]+x[n]

55–75 MIN

A simple recursive smoother with a strong low-frequency preference.

$$y[n] - 0.5y[n-1] = x[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$

$$h[n] = (0.5)^n u[n]$$

At $\omega=0$

- $H=1/(1-0.5)=2$
- DC is amplified

At $\omega=\pi$

- $H=1/(1+0.5)=2/3$
- Nyquist-frequency alternation is attenuated

Think–Pair–Share: sketch $|H(e^{j\omega})|$ before computing

55–75 MIN

Reason from a few strategically chosen frequencies.

Given $y[n] = 0.5y[n-1] + x[n]$

- 1 minute: derive $H(e^{j\omega})$.
- 2 minutes: evaluate $|H|$ at $\omega=0, \pi/2, \pi$.
- 2 minutes: sketch a smooth 2π-periodic magnitude curve.
- Pair discussion: what audio behavior would you expect?

Prompt

What feature of the recursion causes “memory” of prior samples?

Graduate extension

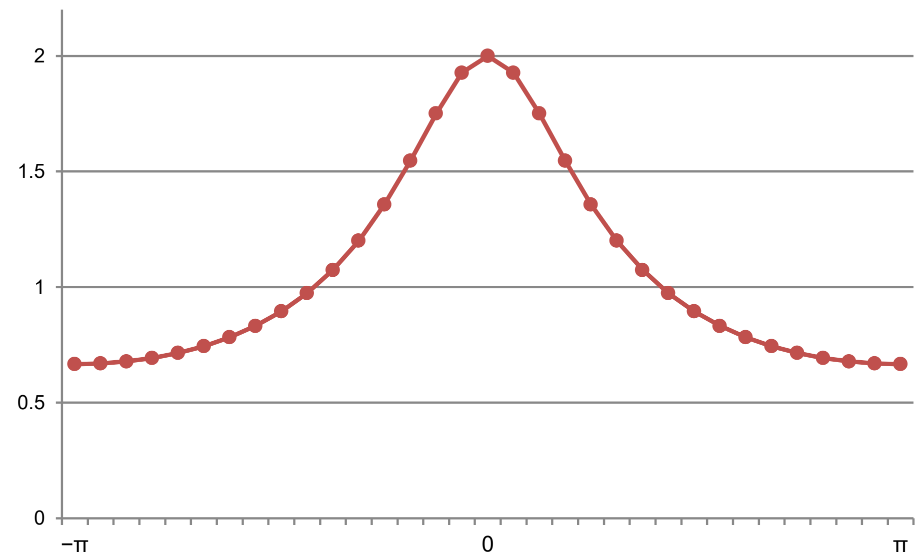
Relate the coefficient 0.5 to a pole at $z=0.5$ and explain why moving it toward 1 increases low-frequency selectivity.

Think–Pair–Share solution

55–75 MIN

The denominator is smallest near ω=0, so the response peaks at low frequency.

$$|H(e^{j\omega})| = \frac{1}{\sqrt{1.25 - \cos\omega}}$$



Three checkpoints

ω=0: 2 • ω=π/2: 0.894 • ω=π: 0.667

Interpretation

Smooth low-pass tendency • recursive memory • stable because |0.5|<1

Python demo: freqz and filtering

55–75 MIN

The numerical tools should confirm—not replace—the analysis.

SciPy frequency response

```
import numpy as np
from scipy import signal

b = [1.0]
a = [1.0, -0.5]
w, H = signal.freqz(b, a, worN=1024)
mag = np.abs(H)
```

Filtering

```
x = np.random.randn(48000)
y = signal.lfilter(b, a, x)
# Compare spectra / listen if desired
```

Notebook task

Change the recursion coefficient from 0.5 → 0.9 → 0.99. Predict the shape first; then verify with freqz.

DTFT vs DFT vs FFT

55–75 MIN

Three related terms that students often collapse into one.

DTFT

- Input: discrete-time sequence
- Frequency: continuous ω
- 2π-periodic spectrum
- Mathematical transform

DFT / FFT

- Input: finite N samples
- Frequency: N discrete bins
- DFT is the transform
- FFT is an algorithm for the DFT

Practical bridge

In code, we sample the DTFT-like spectrum on a finite grid using the DFT/FFT. The grid and window matter.

ML bridge: spectrograms are structured frequency-domain features

BRIDGE

The DTFT is the conceptual ancestor of the STFT.

Waveform — x[n] ❯ **Window** — local segment ❯ **DTFT/DFT** — local spectrum ❯ **Stack over time** — spectrogram ❯ **CNN** — learn patterns

- DSP explains what the axes and magnitudes mean.
- Windowing and sampling choices determine resolution and leakage.
- ML operates on features whose structure comes directly from signal-processing choices.

Concept check

CHECK

Reason from the denominator, not from memorized filter labels.

For $H(e^{j\omega}) = 1/(1 - 0.9e^{-j\omega})$, where is $|H|$ largest?

A. Near $\omega=0$

B. Near $\omega=\pi/2$

C. Near $\omega=\pi$

D. It is constant for all ω

WRAP-UP

Lecture 2 synthesis

The frequency domain turns LTI systems into multiplication.

Complex exponential	Eigenvalue	DTFT	Convolution theorem	Spectral features
e^{jωn}	H(e^{jω})	X(e^{jω})	Y=XH	energy / shape

- Be able to derive H(e^{jω}) from h[n] or a difference equation.
- Be able to explain 2π periodicity and the relation ω=2πf/fs.
- Be able to reason qualitatively about a response before plotting it.

Week 2

Sampling will explain which analog frequencies map to the same discrete-time frequency—and why aliasing occurs.

Weekly quiz

ASSESSMENT

15 minutes • closed notes unless your course policy says otherwise

- Core skills: LTI properties, convolution, eigenfunction reasoning, DTFT periodicity.
- One short calculation: derive $H(e^{j\omega})$ from a first-order difference equation.
- One conceptual explanation: interpret a frequency response without plotting.

Suggested use

Give the quiz at the start of Week 2 or as the last 15 minutes of Lecture 2, depending on pacing.